In [1]:
```
import matplotlib.pyplot as plt
from sklearn.datasets import load_digits
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
```

In [2]:
```
digits=load_digits()
X=digits.data
print(X)
```

```
[[ 0.  0.  5. ...  0.  0.  0.]
 [ 0.  0.  0. ... 10.  0.  0.]
 [ 0.  0.  0. ... 16.  9.  0.]
 ...
 [ 0.  0.  1. ...  6.  0.  0.]
 [ 0.  0.  2. ... 12.  0.  0.]
 [ 0.  0. 10. ... 12.  1.  0.]]
```

In [3]:
```
k_values=[5,10,15,20]
inertia=[]
silhouette=[]
for k in k_values:
    kmeans=KMeans(n_clusters=k,random_state=42,n_init=10)
    kmeans_labels=kmeans.fit_predict(X)
    inertia.append(kmeans.inertia_)
    score=silhouette_score(X,kmeans_labels)
    silhouette.append(score)

    print("K:",k)
    print("Inertia:",round(kmeans.inertia_,2))
    print("Silhouette Score:",round(score,3))
    print("\n----------------------")
```

```
K: 5
Inertia: 1497730.42
Silhouette Score: 0.138

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K: 10
Inertia: 1165725.11
Silhouette Score: 0.182

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K: 15
Inertia: 1019436.35
Silhouette Score: 0.186

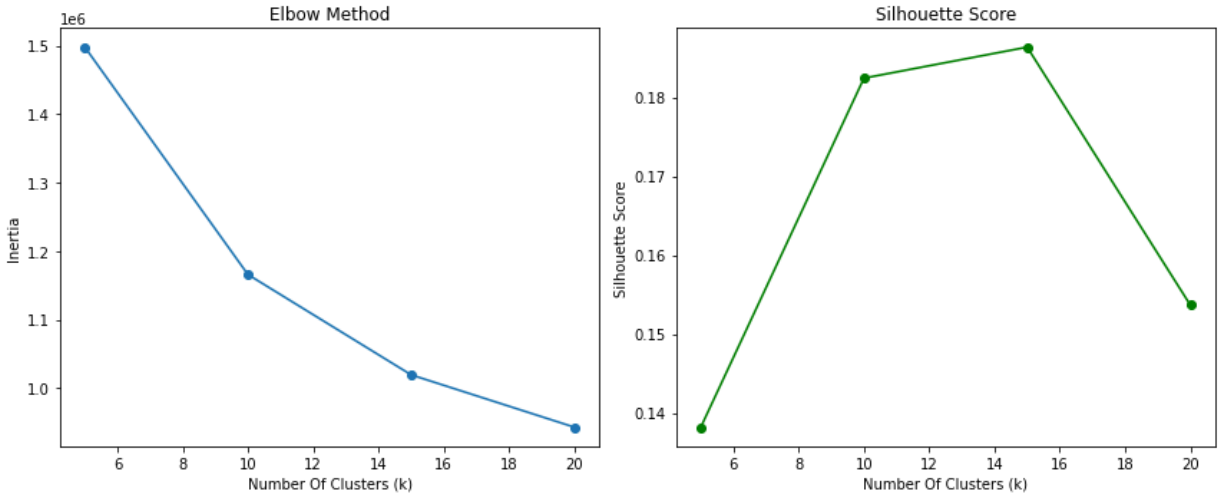
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K: 20
Inertia: 943019.75
Silhouette Score: 0.154

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```

In [4]:
```
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
plt.plot(k_values,inertia,marker='o')
plt.title('Elbow Method')
plt.xlabel('Number Of Clusters (k)')
plt.ylabel('Inertia')

plt.subplot(1,2,2)
plt.plot(k_values,silhouette,marker='o',color='green')
```

```
plt.title('Silhouette Score')
plt.xlabel('Number Of Clusters (k)')
plt.ylabel('Silhouette Score')
plt.tight_layout()
plt.show()
```



In [ ]: